

Chapter 1

Embedded Hardware Design

ENEL712 Embedded Systems Design — Lecture 7

Summary

End-to-end hardware design process for an embedded product, illustrated by a NZ vineyard remote weather station: I/O selection, MCU selection, and power-supply design.

Topics Covered

- Hardware design process
- I/O device selection
- Microcontroller selection
- Power supply design
- Switching vs linear regulators
- Next steps after hardware selection

1.1 The Hardware Design Process

Briefs are usually short and lacking technical detail (e.g., 'Design a remote weather station for NZ vineyards'). Designers must address MCU selection, power supply, components (SMD vs through-hole), PCB design, and interfacing.

First step: draw a block diagram on paper to visualise system components and their connections.

1.1.1 Vineyard weather station block components

- MCU (μC)
- Power supply and battery (LiPO)
- Loads: temperature, humidity, rainfall, sunlight sensors
- Wireless transceiver (T/R)
- Memory (e.g., EEPROM for settings)

- Display (optional, for debugging or local status)
- JTAG/ISP header for programming and debugging

1.2 I/O Device Selection

I/O devices are typically chosen before the MCU so the processor can support all required interfaces.

- Sensors (temperature, light, movement)
- Actuators (motors, solenoids, speakers — less relevant for the weather station)
- User input (buttons, keypads)
- Displays (LCDs, LEDs)
- Communications/storage (SPI, UART, external memory/SD cards)

1.2.1 Wireless module requirements

- Communication range ≥ 1 km
- Data rate ≥ 100 kbps
- Operating in the licence-free ISM band
- UART or SPI interface preferred
- Use a pre-configured module (e.g., ZigBee over UART) over a transceiver IC to avoid antenna/impedance design.
- Prefer digital sensor outputs over analogue for easier accurate reading.

1.3 Microcontroller Selection

1.3.1 I/O Tally

- # digital inputs/outputs
- # analogue inputs/outputs (ADC, DAC)
- # communication interfaces (UARTs, SPIs, I²Cs)
- # timers (PWM and Input Capture channels)

1.3.2 Weather station tally example

- 5× digital outputs (e.g., power-down lines, rainfall container empty signal, battery charger status)
- 3× analogue inputs (rainfall, sunshine, battery voltage)
- 1× UART (wireless ZigBee module)
- 2× I²C (temperature and humidity sensors, possibly shared bus)
- 1× SPI (external EEPROM)

1.3.3 Selected MCU example

ATmega 328P (picoPower): 32 KB Flash, 2 KB RAM, 1.8 V – 5 V operation, picoPower mode for long battery life. Verify all required pins (including ISP) are available on the chosen package.

1.4 Power Supply Design

- DC-to-DC supply must consider input/output voltages, noise sensitivity, current, and efficiency.
- Sleep modes are critical for the battery system — wake on WDT or RTC alarm.
- Build a current tally for the supply, regulators, and all loads.

1.4.1 Current tally example

Rail / total	Voltage	Current
5 V rail total	5 V	101 mA
3.3 V rail total	3.3 V	216 mA
Total (calculated)	—	317 mA
Total (with 20 % margin)	—	380 mA

3.3 V rail: MCU (10 mA), wireless T/R (205 mA), I²C sensors. 5 V rail: rainfall sensor and sunshine sensor.

1.5 Switching vs Linear Regulators

Regulator	Pros	Cons
Switching	High efficiency (>80 %); good for large Vin/Vout differences	Complex to design; introduces electrical noise
Linear	Simple and cheap	Inefficient; dissipates excess as heat

A common hybrid approach uses a switching regulator first (efficiency) followed by a linear regulator and decoupling caps (clean output).

1.6 Next Steps

After hardware selection, move to schematic design (e.g., Altium) following datasheets for tips and required external components. Software planning can begin in parallel — hardware is not required to start coding.

1.7 Use Cases & Applications

- Iterating from brief to schematic: the I/O tally → MCU choice → power tally pipeline applies to almost every battery-powered embedded product.
- Pre-configured wireless modules let small teams ship a wireless product without an RF specialist — the trade-off is per-unit cost.

- Hybrid switching + linear supplies are common in audio, sensor frontends, and any analog circuitry that is sensitive to switching ripple.

1.8 Formulas & Calculations

1.8.1 Battery life from current tally

Lifetime (h) $\approx$ Battery capacity (mAh) $\div$ Average current (mA). For varying loads, average current is the time-weighted mean across all operating modes.

1.8.2 Worked example: vineyard station

If the device wakes for 1 s every 60 s drawing 380 mA, and sleeps the rest at 50 μ A on a 500 mAh battery:

- Active duty cycle: $1 / 60 \approx 1.67 \%$.
- $I_{avg} = 0.0167 \times 380 \text{ mA} + 0.9833 \times 0.05 \text{ mA} \approx 6.39 \text{ mA}$.
- Lifetime $\approx 500 / 6.39 \approx 78 \text{ h} \dots$ so we'd need to either cut active time, lower active current, drop the wake rate, or move to a bigger battery.

1.8.3 Sizing a 20 % current margin

$I_{target} = 1.20 \times \Sigma I_{max}(\text{component})$. Margin covers component tolerance, transient peaks (e.g., Wi-Fi TX bursts), and inrush.

1.9 Design Checklist

- Block diagram drawn before any chip is selected.
- I/O tally signed off against a candidate MCU pinout (and ISP/JTAG header verified).
- Current tally per voltage rail, with margin.
- Sleep state path: which oscillator is running? what wakes us? what is the residual current?
- Schematic captures all manufacturer-recommended decoupling, pull-ups, and crystal load capacitors.

Key Definitions

ADC

Analog-to-Digital Converter; turns continuous voltage into discrete digital values.

Current Tally

Sum of every component's max current (with margin) used to size each voltage rail.

I/O Tally

Comprehensive count of digital, analogue, communication, and timer peripherals required.

ISM Band

Industrial, Scientific, Medical bands (e.g., 900 MHz, 2.4 GHz) — licence-free for embedded wireless.

ISP

In-System Programming header for programming/debugging the MCU on-board.

Linear Regulator

Voltage regulator that drops excess voltage as heat; simple but inefficient.

Sleep Modes

Low-power MCU states that shut down peripherals to drastically reduce current.

Switching Regulator

Inductor-based regulator with high efficiency but more complexity and noise.

picoPower

Atmel marketing term for AVRs with very low active and sleep currents (down to $<1 \mu\text{A}$ in power-down).

LDO Dropout

Minimum $V_{in} - V_{out}$ the regulator needs to keep regulating; below this the output sags.

Decoupling Capacitor

Small capacitor near a chip's power pin that supplies transient currents and filters noise.

Sources

- Lecture 7.pdf